LÂM GIA PHÚC NGUYÊN

COMPUTER SCIENCE STUDENT

Ho Chi Minh City, Vietnam | contact@rawsie.dev | github.com/rawsie-dev | rawsie.dev

EDUCATION

VNU-HCM High School for the Gifted (PTNK) **August 2024 – June 2027 (Expected)**

High School Diploma, Informatics Specialization

- **GPA:** 9.7/10 (Grade 9, previous school); 9.6/10 (Grade 10); 9.6/10 (Grade 11)
- **Standardized testing:** SAT: 1520 | IELTS: 8.0 | AP Calculus AB: 5, AP Calculus BC: 5, AP Computer Science A: 5, AP Computer Science Principles: 4.
- **Relevant coursework:** Machine Learning, Deep Learning, Data Structures & Algorithms.

RESEARCH & ACADEMIC EXPERIENCE

Brain-Inspired Neural Networks **June 2026 – Present**

Exploring Brain-Inspired Functional Diversity and Competitive Sparsity in Artificial Neural Networks

First author

- Investigated functional diversity and competitive sparsity as brain-inspired approaches to improving ANNs.
- Designed and evaluated parameter-free neural network modifications through controlled experiments on tabular and image classification tasks.
- Analyzed their effects on feature redundancy and generalization without introducing additional trainable parameters; authored a first-author research manuscript [View paper].

Synthetic Image Detection Research **June 2025 – October 2025**

Mattaru Net: Lightweight synthetic images detection using contrastive distilled learning

Co-author

- Conducted research on lightweight detection of AI-generated images as part of the MediaEval 2025 campaign.
- Built core data-processing and evaluation pipelines, analyzed the dataset, and led experimental runs.
- Co-authored a published study on lightweight synthetic-image detection using distilled contrastive representations; **published** in the MediaEval 2025 proceedings [View paper].

INTERNSHIP EXPERIENCE

BOCC - Invoice Information Extractor **May 2026 – Present**

Software Engineering Intern

- Identified opportunities to automate manual invoice processing and independently developed an offline desktop application to extract structured information from Vietnamese invoices.
- Integrated PaddleOCR, VietOCR, and optional local LLM inference into a PySide6 application; developed its interface and packaging workflow.
- Enabled local invoice processing without cloud-based inference, supporting data privacy and reducing manual information entry.

EXTRACURRICULAR EXPERIENCE & LEADERSHIP

The Gifted Battlefield (Student Organization)

A student-led academic platform providing free and specialized study materials for gifted middle and high school students throughout the country.

Vice Head of Informatics Department **August 2025 – Present**

- Co-led an 18-member team overseeing academic activities, mock examinations, and informatics competitions.
- Developed and reviewed specialized competitive programming materials for regional competitions and mock examinations serving 50+ high school participants.

Member, Informatics Department **August 2024 – August 2025**

- Contributed to academic initiatives and supported the preparation of competitive programming materials and mock examinations.

Vo Uu Pagoda (Chùa Vô Ưu)

Volunteer, Mathematics Instructor (Grade 8) **June 2025 – August 2025 & June 2026 – August 2026**

- Volunteered to teach mathematics to Grade 8 students during two consecutive summer terms.
- Designed and prepared math lessons, assignments, tests, and exams customized to students' learning needs and taught two cohorts of eight students.

Informatics Major's Traditional Festival

Vice Head of Events Committee **June 2025 – February 2026**

- Helped organize the traditional festival for the Informatics major.
- Pitched festival games and a stamp-and-reward system; helped to write the MC script and an event proposal; staffed a station on the day of the event.

Red Flamboyant Flower Summer Volunteer Campaign (Hoa Phượng Đỏ) 2026

Member, "Digital Mass Literacy" Unit **June 2026 – September 2026**

- Participated in community digital-literacy activities supporting access to essential digital services.
- Developed supporting materials, including a presentation on essential Vietnamese apps for older adults, and assisted with in-person activities.

PiMA 2026 MATHEMATICS AND APPLICATIONS SUMMER CAMP

Camper **August 2026**

- Attended a summer camp exploring applications of mathematics across science, technology, and AI.
- Participated in lectures and collaborative problem-solving; applied Hidden Markov Models to predict gene function from histone marks as part of a group research project.

PROJECTS

Personal Website **2026**

Developed a personal website showcasing my background, projects, and writing using SvelteKit and MDsveX [rawsie.dev].

Piano Music Visualizer **2026**

Built a browser-based falling-note visualizer for original piano compositions using SvelteKit, TypeScript, Tone.js, and MIDI [piano.rawsie.dev].

ACHIEVEMENTS

- **National Round Qualifier**, Vietnam Olympiad in Artificial Intelligence (VOAI), 2026
  - Ranked among the top 246 of 1,188 participants in the national qualifying round, advancing toward selection for Vietnam's international delegation.
- **Top 8 (Consolation Prize)**, PTNK Program in Interdisciplinary Sciences & Education (PIISE) 2025, a school-level science and innovation competition; exhibited research at the competition.
- **Consolation Prize**, National Young Informatics Competition (*Tin Học Trẻ*), 2024
- **Third Prize**, City-Level Excellent Student Competition in English, 2023
- **First Prize**, City-Level Young Informatics Competition (*Tin Học Trẻ*), 2020

TECHNICAL SKILLS

- **Programming:** Python, C++, JavaScript
- **Web Technologies:** HTML, CSS, SvelteKit, TypeScript
- **Frameworks & Libraries:** PyTorch, TensorFlow (Keras), PaddleOCR, VietOCR, PySide6
- **Technical Areas:** Machine Learning, Computer Vision, Data Structures & Algorithms

ADDITIONAL INFORMATION

- **Languages:**
  - Vietnamese (Native)
  - English (Fluent)
- **Interests:**
  - Machine Learning Research
  - Competitive Programming
  - Piano Performance and Composition